Current Affairs 12-11-2026

Credit - Drishti Ias

Cloud Seeding

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Source: IE

Why in News?

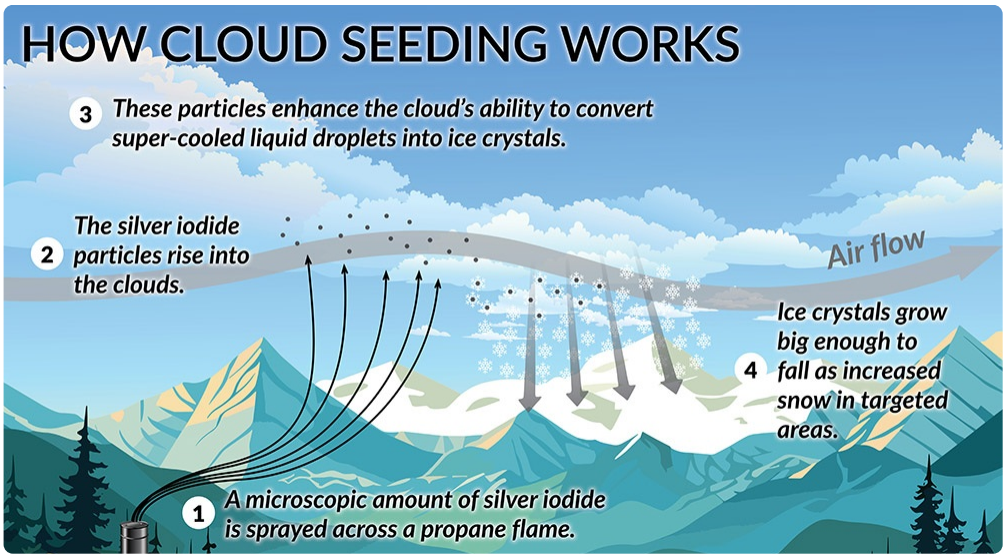
As Delhi and the National Capital Region (NCR) face hazardous **Air Quality Index (AQI)** levels post-Diwali in November 2026, authorities have once again proposed **Cloud Seeding (Artificial Rain) as an emergency mitigation strategy** .

- However, the efficacy of this attempt is being questioned, as the **previous trials conducted in October 2025** yielded negligible rainfall and failed to create any tangible, lasting improvement in the capital's air quality.

What is Cloud Seeding (Artificial Rain)?

- **About:** Cloud seeding is a weather modification technology that **artificially induces or enhances precipitation** by introducing specific chemical substances (Seeding Agents) into existing cloud formations.
  - It aims to accelerate the **natural rain-making process** to manage drought, combat severe air pollution, or suppress hail.
- **Types:**
  - **Glaciogenic Seeding:** Uses ice-nucleating agents such as silver iodide (AgI) to promote ice-crystal formation in suitable cold or supercooled clouds, enhancing snowfall or rainfall.
  - **Hygroscopic Seeding:** Introduces moisture-attracting particles such as **sodium chloride (NaCl)** into suitable warm clouds to encourage droplet growth through collision and coalescence, potentially increasing rainfall.
- **Mechanism of Action:**
  - **Naturally:** Water vapour condenses onto **cloud condensation nuclei (CCN),** such as dust, sea salt, and smoke particles, forming tiny cloud droplets.
    - These tiny droplets grow through the condensation of additional water vapour and collision and coalescence with other droplets.
    - When droplets become sufficiently large and heavy, they can no **longer remain suspended in the atmosphere and fall as precipitation (rain).**
    - In colder clouds, water droplets may freeze or water vapour may **deposit directly onto ice nuclei, forming ice crystals** . These crystals grow and **eventually fall as snow or melt into rain.**

- **In Cloud Seeding:** Dispersed chemical agents (delivered via aircraft, drones, or ground generators) act as **artificial Cloud Condensation Nuclei (CCN) or ice nuclei.**
  - They provide a **dense, highly effective base** for water vapor to condense and freeze around, **accelerating the formation of water droplets** or ice crystals. Once these droplets become heavy enough, they fall to the ground as artificial rain or snow.



- **Impact on Air Pollution:**
  - **Washout Effect (Coagulation):** As artificial rain falls, the droplets attract and intercept suspended particulate matter (**PM2.5** and **PM10**), soot, sulfates, and toxic **aerosols.**
  - **Settling:** The precipitation physically washes these pollutants out of the atmosphere and deposits them on the ground, leading to a temporary but rapid improvement in the **AQI and surface visibility.**
- **Meteorological Conditions:**
  - **Pre-existing Clouds:** Cloud seeding cannot generate rain from clear skies; it strictly requires existing cloud cover.
  - **Cloud Depth and Moisture:** The target clouds must have sufficient vertical depth and adequate **liquid water content (typically 15-20% humidity in the targeted cloud layer).**
  - **Synoptic Systems:** In regions like North India during winter, cloud seeding relies on the presence of **Western Disturbances** —extratropical storms originating in the **Mediterranean** that bring necessary moisture and cloud formations.
- **Limitations and Challenges of Cloud Seeding:**
  - **Short-Term Relief:** It only addresses the symptoms of air pollution, not the source. An IIT Delhi study found that even after substantial natural rainfall, **pollutant concentrations generally return to pre-rainfall levels within 1–5 days due to continued emissions** .
    - It also highlighted a possible increase in **ground-level ozone concentrations following rainfall.**
  - **Meteorological Dependency:** Success is entirely contingent on unpredictable weather variables.
    - Over Delhi, **winter clouds are often too dry** or shallow to yield significant rainfall even when seeded.
  - **High Capital Cost:** Deploying specialized aircraft equipped with **pyrotechnic flares** and aviation fuel is highly resource-intensive (costing crores for a limited coverage area)
- **Ecological and Ethical Concerns:**
  - **Toxicity and Bioaccumulation:** Prolonged use of silver iodide can lead to the **accumulation of**

heavy metals in local soils and water bodies, potentially harming aquatic life and agricultural ecosystems.

- **Rain Theft:** Altering weather patterns in one micro-region might artificially **deprive neighboring, naturally drought-prone areas of their expected precipitation.**
- **Carbon Footprint:** Using **dry ice (carbon dioxide)** and flying aviation-fuel-heavy aircraft directly contribute to **greenhouse gas emissions.**

Global Examples of Cloud Seeding Programmes

- **United Arab Emirates (UAE):** Since the 1990s, the UAE has conducted around 1,000 hours of cloud-seeding flights annually to enhance desert rainfall and replenish groundwater.
- **Australia:** Cloud seeding is used to augment water supplies and hydropower generation, particularly in Tasmania and the Snowy Mountains.
- **United States:** Western states such as California, Colorado and Texas use cloud seeding to increase mountain snowpack, supporting reservoirs, irrigation and drought management.
- **China:** Operates one of the world's largest weather-modification programmes using aircraft, drones and artillery for artificial rainfall, drought relief and hail suppression

Alternative Solutions (Beyond Cloud Seeding):

- **Wind Augmentation Purifying Unit (WAYU):** Developed by the Council of Scientific and Industrial Research - National Environmental Engineering Research Institute (**CSIR-NEERI**), these devices are deployed at high-traffic intersections.
  - They work on the principles of **wind generation** and active pollutant removal, utilizing particulate filters and activated carbon/UV systems to neutralize **Carbon Monoxide (CO) and Volatile Organic Compounds (VOCs).**
- **Smog Towers and Downdraft Filtration:** Large-scale outdoor air purification structures (like those tested in Connaught Place, Delhi) that employ downdraft or updraft airflow technology.
  - They pull in polluted air from the top, pass it through thousands of macro and **micro- HEPA (High-Efficiency Particulate Air) or electrostatic filters,** and release clean air at the bottom to create localized clean air zones.
- **Photocatalytic Infrastructure (Smog-Eating Paints):** The application of paints or building coatings infused with **Titanium Dioxide (TiO2)** on urban infrastructure.
  - When exposed to sunlight (UV radiation), these photocatalytic materials trigger a chemical reaction that breaks down **toxic Nitrogen Oxides (NOx) from vehicle exhaust into harmless nitrate salts,** which are then washed away by natural rain.
- **Anti-Smog Guns and Mist Cannons:** A targeted, ground-level technological intervention that **mimics the washout effect of artificial rain.**
  - These machines spray atomized water droplets **(typically 50-100 microns in size) into the air at high velocities to coagulate** and settle suspended particulate matter (PM2.5 and PM10) around construction sites and mining areas.
- **Pusa Decomposer:** A highly effective biotech alternative to agricultural fires, developed by the **ICAR** .

- It is a **microbial consortium** (fungi-based bio-enzyme liquid) sprayed onto harvested fields that rapidly accelerates the decomposition of paddy stubble, turning it into organic manure within 15-20 days without the need for burning.

- **AI-Driven Smart Traffic Management:** Implementing **Artificial Intelligence (AI) and Internet of Things (IoT)** synchronized traffic grids to eliminate vehicle idling at red lights.
  - By optimizing traffic flow through real-time data, urban centers can drastically cut down localised concentrations of **tailpipe emissions (NOx and Carbon Monoxide)** at major junctions.
- **Urban Greening:** Developing dense urban forests (**Miyawaki method**) and green corridors to act as natural dust sinks.
- **In-Situ Stubble Management:** Subsidizing and deploying machinery like **Happy Seeders** and Super SMS to prevent agricultural biomass burning in Punjab, Haryana, and UP.
- **Clean Mobility Transition:** Accelerating the shift to **Electric Vehicles (EVs)** through government subsidies and expanding public transit networks to reduce tailpipe emissions.

Frequently Asked Questions (FAQs)

1. What is cloud seeding and which substances are commonly used?
Cloud seeding is a weather modification technique that enhances precipitation by introducing seeding agents such as silver iodide, potassium iodide, sodium chloride and dry ice into existing clouds.

2. What are Cloud Condensation Nuclei (CCN)?
CCNs are tiny atmospheric particles, such as dust, sea salt and smoke, that provide surfaces for water vapour to condense and form cloud droplets.

3. Why is cloud seeding dependent on Western Disturbances in Delhi-NCR?
Western Disturbances bring moisture-bearing clouds and favourable atmospheric conditions during winter, potentially supporting cloud-seeding operations. However, cloud seeding cannot produce rainfall in clear skies.

4. What is the role of CSIR-NEERI's WAYU in controlling air pollution?
The Wind Augmentation Purifying Unit (WAYU) uses wind generation and pollutant-removal mechanisms to reduce local concentrations of particulate matter and gaseous pollutants, particularly at traffic intersections.

5. How does the Pusa Decomposer help control air pollution?
Developed by ICAR, Pusa Decomposer is a microbial consortium that accelerates the decomposition of paddy stubble into organic manure, providing an alternative to crop residue burning.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/86MozHiK7zE**]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. Artificial way of causing rainfall to reduce air pollution makes use of (2025)

(A) silver iodide and potassium iodide

(B) silver nitrate and potassium iodide

(C) silver iodide and potassium nitrate

(D) silver nitrate and potassium chloride

Ans: A

Q. In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosol into the stratosphere? (2019)

(a) Creating the artificial rains in some regions

(b) Reducing the frequency and intensity of tropical cyclones

(c) Reducing the adverse effects of solar wind on the Earth

(d) Reducing the global warming

Ans: (d)

Siddis: Gujarat's African-Origin Tribal Community

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Source: IE

The historic achievement of **Khushbu Rajakbhai Darjada,** a 15-year-old judoka who became the **first Siddi athlete to win a Youth Olympics medal** and only the **second Indian judoka** to achieve this feat, has brought renewed attention to the Siddi community, a tribal community of African descent primarily residing in Gujarat.

- **Origins and Arrival:** The Siddis trace their genetic ancestry to **East Africa** .
  - There is no single definitive record of their arrival in India. Some historical accounts (like the **1931 Census**) suggest they were brought as slaves by **Portuguese colonists** in the 17th century, or by Central/West Asian invaders in the 15th century.
  - Community folklore suggests they were brought over 700 years ago by the **Nawabs of Junagadh** to serve as bodyguards (*angrakshaks*), strongmen, and caretakers of captive lions.
- **Demographics and Geography:**
  - **Population:** They are a small community, as per a 2025 government survey, the population of **Siddis in Gujarat classified as a Particularly Vulnerable Tribal Group (PVTG) is 12,959** .

- **Location:** Their population in Gujarat is heavily concentrated in the Saurashtra region, specifically in 18 villages of **Talala taluka in Gir Somnath district** . Jambur village is often referred to as **"Mini Africa."**
    - Members also reside in Maharashtra, Goa, Diu, and Daman.
- **Cultural Assimilation:** Despite their African physical features, Siddis are entirely assimilated into Gujarati culture. They speak Gujarati with a Kathiawadi accent, eat local food, and dress like native Gujaratis.
    - They are predominantly **Muslim** , with their religious beliefs heavily influenced by Sufism, while also incorporating some regional Hindu elements.
    - They revere **Baba Gor** , a 15th- or 16th-century African Muslim saint.
- **Constitutional Status:** Siddis residing in the **Saurashtra region** were declared a **Scheduled Tribe (ST)** in 1956.
    - In 1982, the Government of India classified them as a **PVTG** (formerly known as a "primitive tribal group").
    - However, Siddis living outside the **Saurashtra region** (e.g., in Bharuch) are not classified as STs; they fall under the **Socially and Economically Backward Community (SEBC) category.**
- **Livelihood and Lifestyle:** Due to their proximity to the **Gir National Park** , many Siddis work as forest guides, drivers, and laborers.
    - Often called *Siddi Badshah* (carefree kings), they are known for their simple, free-spirited nature.
    - **Dance:** The Siddis are known for Dhamal, a traditional, energetic folk dance of African origin, performed by men and featuring animal-like movements and calls. Women participate in other traditional Siddi dances.
- **Sporting Prowess:** Considered natural athletes, many Siddi youths excel in sports, particularly martial arts and athletics, leading to their selection in national sports camps, though adapting to strict discipline remains a cultural challenge for some.



Read more: Siddi Community